HW1 – Read Me

Ex 1 – 2D “Top - Down Shooter” Game

The game consists of a spaceship, which has to shoot the asteroids coming towards it.

There are 3 levels of difficulty: the number and the size of the asteroids increase, as well as their speed and the number of bullets it takes to destroy each of them. Bigger asteroids have a higher score and need more bullets to be destroyed. If the spaceship destroys an asteroid, its score is gained. If an asteroid hits the spaceship, its score is subtracted from the health score (HP). All this information is shown during the game as messages in the console.

The player wins the game if the spaceship shoots down all the asteroids through all the levels and loses when the HP is less than zero.

For creating the game, we used modern OpenGL and C++.

Main.cpp file:

1. *cursor_position_callback* function - to translate the cursor positions into viewport points.
2. *mouse_button_callback* function - to check if the mouse buttons are clicked and what happens when the user clicks either the left or the right one. If the user clicks the left button, it lets the program know to shot a bullet in it direction and if the user click right the game starts or the difficulty level is increased.
3. The *main* function. We have 3 programId’s for: spaceship, bullets and asteroids.
    - First, we check the level and we change the color of the background accordingly (level 1: blue, level 2: green, level 3: orange).
    - We set the WASD keys for the ship’s movement.
    - We check if the health is higher than 0.
    - After the bullet is shot (clicks left), the bullet is translated to the cursor position, then is translated back inside the spaceship and the clicking index goes back to 1. The index will have the value 0 at the next click.
    - If an asteroid gets shot, then the user gains the score for this asteroid (+100, +200 or +300 points) depending on the asteroid’s level.
    - In case an asteroid hits the spaceship, the score for this asteroid is subtracted (-100, -200 or - 300 points) from the HP and the asteroid and the spaceship are translated in opposite directions.
    - All the messages about the score are shown in the console.
    - After finishing the game, the user can click right to restart.

Ex 2 – 3D “Petanque” Game

The game consists of a ground plane, 6 balls with 3 for each team and a small ball, the jack. For creating the game, we used modern OpenGL and C++.

1. Keys that are used in the game:
    - ***V*** - spawns the balls on the ground from outside the frame.
    - ***P*** -launches the ball towards the jack.
    - ***Left and Right arrow keys*** – moves the ball to position it before the launch.
    - ***WSRF keys*** – translate the camera front, back, up and down.
    - **UHJK keys** – rotate the camera.
    - **O**- resets the game.
2. The *main* function. We have 3 programId’s for: balls, jack and the ground plane.
    - Checks the position and the state of a ball, then launches it to a random position near the jack. Checks the color and spawns the alternatively.
    - After all 6 balls are launched, the score for each team is calculated by adding the distance of each ball to the jack. The team with the smaller sum wins the game.